

How do screens make **EVERY** color?

PIXEL = Picture Element

Every pixel = R + G + B sub-pixels

Adding LIGHT makes new colors

Paint absorbs light — Screens emit it

Each sub-pixel: 0 → 255 (256 levels)

$256^3 \approx \underline{16.7 \text{ MILLION}}$ colors

All ON = White • All OFF = Black

Your eyes BLEND tiny sub-pixels

LCD = backlight • OLED = self-lit pixels

3 colors → millions of shades

1. The 3 primary screen colors? 2. Brightness range of a sub-pixel? 3. Red + Green light = ?

**1. Red, Green, Blue (RGB) 2. 0 to 255
(256 levels) 3. Yellow**

3 colors. Millions of shades. SUBSCRIBE!